

Ch8

Plane Electromagnetic Waves

Ch8: Plane Electromagnetic Waves:

Plane waves in lossy media

P.8-1

Write down Maxwell's Equations and obtain the wave equations governing the $\mathbf{E}$ and $\mathbf{H}$ fields in a source-free conducting medium with constitutive parameters ϵ , μ and σ .

(Hint: $\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$)

Solution

In a source-free conducting medium with constitutive parameters ϵ, μ and σ ,

$$\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t} \quad - \textcircled{1}$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

$$\mathbf{J} = \mathbf{J}_s + \mathbf{J}_c$$

$\mathbf{J}_s$: Source current density $\therefore$ source-free $\therefore \mathbf{J}_s = \mathbf{0}$

$\mathbf{J}_c$: Conduction current density $\mathbf{J}_c = \sigma \mathbf{E}$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} = \sigma \mathbf{E} + \epsilon \frac{\partial \mathbf{E}}{\partial t} \quad - \textcircled{2}$$

Solution

Take the curl of Maxwell's Equations,

$$\nabla \times (\nabla \times \mathbf{E}) = -\mu \frac{\partial}{\partial t} (\nabla \times \mathbf{H})$$

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}$$

$$\because \nabla \cdot \mathbf{D} = \nabla \cdot (\epsilon \mathbf{E}) = 0$$

$$\therefore \nabla \cdot \mathbf{E} = 0$$

$$\nabla \times (\nabla \times \mathbf{E}) = -\nabla^2 \mathbf{E} = -\mu \frac{\partial}{\partial t} (\nabla \times \mathbf{H}) \quad - \textcircled{3}$$

Solution

Combine ② and ③,

$$\nabla \times (\nabla \times \mathbf{E}) = -\nabla^2 \mathbf{E} = -\mu \frac{\partial}{\partial t} (\nabla \times \mathbf{H})$$

$$\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t} \quad - \text{①}$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} = \sigma \mathbf{E} + \varepsilon \frac{\partial \mathbf{E}}{\partial t} \quad - \text{②}$$

$$-\nabla^2 \mathbf{E} = -\mu \frac{\partial}{\partial t} (\nabla \times \mathbf{H}) = -\mu \frac{\partial}{\partial t} \left(\sigma \mathbf{E} + \varepsilon \frac{\partial \mathbf{E}}{\partial t} \right)$$

Thus,

$$\nabla^2 \mathbf{E} - \mu \frac{\partial}{\partial t} \left(\sigma \mathbf{E} + \varepsilon \frac{\partial \mathbf{E}}{\partial t} \right) = 0$$

$$\nabla^2 \mathbf{E} - \mu \sigma \frac{\partial \mathbf{E}}{\partial t} - \mu \varepsilon \frac{\partial^2 \mathbf{E}}{\partial t^2} = 0$$

Solution

Similarly for H ,

$$\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t} \quad - \textcircled{1}$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} = \sigma \mathbf{E} + \varepsilon \frac{\partial \mathbf{E}}{\partial t} \quad - \textcircled{2}$$

$$\nabla \times (\nabla \times \mathbf{H}) = \sigma(\nabla \times \mathbf{E}) + \varepsilon \frac{\partial}{\partial t} (\nabla \times \mathbf{E})$$

$$\nabla \times (\nabla \times \mathbf{H}) = \nabla(\nabla \cdot \mathbf{H}) - \nabla^2 \mathbf{H}$$

$$\because \nabla \cdot \mathbf{B} = \nabla \cdot (\mu \mathbf{H}) = 0$$

$$\therefore \nabla \cdot \mathbf{H} = 0$$

$$\nabla \times (\nabla \times \mathbf{H}) = -\nabla^2 \mathbf{H} = \sigma(\nabla \times \mathbf{E}) + \varepsilon \frac{\partial}{\partial t} (\nabla \times \mathbf{E}) \quad - \textcircled{4}$$

Solution

Combine ① and ④,

$$\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t} \quad - \text{①}$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} = \sigma \mathbf{E} + \varepsilon \frac{\partial \mathbf{E}}{\partial t} \quad - \text{②}$$

$$\nabla \times (\nabla \times \mathbf{H}) = -\nabla^2 \mathbf{H} = \sigma(\nabla \times \mathbf{E}) + \varepsilon \frac{\partial}{\partial t} (\nabla \times \mathbf{E})$$

$$-\nabla^2 \mathbf{H} = \sigma(\nabla \times \mathbf{E}) + \varepsilon \frac{\partial}{\partial t} (\nabla \times \mathbf{E}) = -\mu\sigma \frac{\partial \mathbf{H}}{\partial t} - \mu\varepsilon \frac{\partial^2 \mathbf{H}}{\partial t^2}$$

Thus,

$$\nabla^2 \mathbf{H} - \mu\sigma \frac{\partial \mathbf{H}}{\partial t} - \mu\varepsilon \frac{\partial^2 \mathbf{H}}{\partial t^2} = 0$$

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P.8-4

For a harmonic uniform plane wave propagating in a simple medium, both $\mathbf{E}$ and $\mathbf{H}$ vary in accordance with the factor $\exp(-j\mathbf{k} \cdot \mathbf{R})$ as indicated in Eq. (8-26). Show that the four Maxwell's equations for uniform plane wave in a source-free region reduce to the following:

$$\mathbf{k} \times \mathbf{E} = \omega\mu\mathbf{H},$$

$$\mathbf{k} \times \mathbf{H} = -\omega\epsilon\mathbf{E},$$

$$\mathbf{k} \cdot \mathbf{E} = 0,$$

$$\mathbf{k} \cdot \mathbf{H} = 0.$$

Solution

$$\mathbf{E}(\mathbf{R}) = \mathbf{E}_0 e^{-j\mathbf{k}\cdot\mathbf{R}}, \quad \mathbf{H}(\mathbf{R}) = \mathbf{H}_0 e^{-j\mathbf{k}\cdot\mathbf{R}}$$

Take Cartesian Coordinate System as an example,

$$\begin{aligned}\nabla(e^{-j\mathbf{k}\cdot\mathbf{R}}) &= (\mathbf{a}_x \frac{\partial}{\partial x} + \mathbf{a}_y \frac{\partial}{\partial y} + \mathbf{a}_z \frac{\partial}{\partial z}) e^{-(jk_x x + jk_y y + jk_z z)} = -j(\mathbf{a}_x k_x + \mathbf{a}_y k_y + \mathbf{a}_z k_z) e^{-(jk_x x + jk_y y + jk_z z)} \\ &= -j\mathbf{k} e^{-j\mathbf{k}\cdot\mathbf{R}}\end{aligned}$$

$$\begin{aligned}\nabla \times \mathbf{E} &= \nabla \times (\mathbf{E}_0 e^{-j\mathbf{k}\cdot\mathbf{R}}) = \nabla(e^{-j\mathbf{k}\cdot\mathbf{R}}) \times \mathbf{E}_0 + e^{-j\mathbf{k}\cdot\mathbf{R}}(\nabla \times \mathbf{E}_0) = \nabla(e^{-j\mathbf{k}\cdot\mathbf{R}}) \times \mathbf{E}_0 \quad (\nabla \times \mathbf{E}_0 = 0) \\ &= -j\mathbf{k} e^{-j\mathbf{k}\cdot\mathbf{R}} \times \mathbf{E}_0 = -j\mathbf{k} \times \mathbf{E} = -j\omega\mu\mathbf{H} \Rightarrow \mathbf{k} \times \mathbf{E} = \omega\mu\mathbf{H}\end{aligned}$$

$$\begin{aligned}\nabla \times \mathbf{H} &= \nabla \times (\mathbf{H}_0 e^{-j\mathbf{k}\cdot\mathbf{R}}) = \nabla(e^{-j\mathbf{k}\cdot\mathbf{R}}) \times \mathbf{H}_0 + e^{-j\mathbf{k}\cdot\mathbf{R}}(\nabla \times \mathbf{H}_0) = \nabla(e^{-j\mathbf{k}\cdot\mathbf{R}}) \times \mathbf{H}_0 \quad (\nabla \times \mathbf{H}_0 = 0, \mathbf{J} = 0) \\ &= -j\mathbf{k} e^{-j\mathbf{k}\cdot\mathbf{R}} \times \mathbf{H}_0 = -j\mathbf{k} \times \mathbf{H} = j\omega\varepsilon\mathbf{E} \Rightarrow \mathbf{k} \times \mathbf{H} = -\omega\varepsilon\mathbf{E}\end{aligned}$$

Solution

$$\begin{aligned}\nabla \cdot \mathbf{D} = \nabla \cdot (\varepsilon \mathbf{E}) = 0 &\Rightarrow \nabla \cdot \mathbf{E} = \nabla(e^{-j\mathbf{k} \cdot \mathbf{R}}) \cdot \mathbf{E}_0 + e^{-j\mathbf{k} \cdot \mathbf{R}}(\nabla \cdot \mathbf{E}_0) = \nabla(e^{-j\mathbf{k} \cdot \mathbf{R}}) \cdot \mathbf{E}_0 & (\nabla \cdot \mathbf{E}_0 = 0, \rho = 0) \\ &= -j\mathbf{k}e^{-j\mathbf{k} \cdot \mathbf{R}} \cdot \mathbf{E}_0 = -j\mathbf{k} \cdot \mathbf{E} = 0 \Rightarrow \mathbf{k} \cdot \mathbf{E} = 0\end{aligned}$$

$$\begin{aligned}\nabla \cdot \mathbf{B} = \nabla \cdot (\mu \mathbf{H}) = 0 &\Rightarrow \nabla \cdot \mathbf{H} = \nabla(e^{-j\mathbf{k} \cdot \mathbf{R}}) \cdot \mathbf{H}_0 + e^{-j\mathbf{k} \cdot \mathbf{R}}(\nabla \cdot \mathbf{H}_0) = \nabla(e^{-j\mathbf{k} \cdot \mathbf{R}}) \cdot \mathbf{H}_0 & (\nabla \cdot \mathbf{H}_0 = 0) \\ &= -j\mathbf{k}e^{-j\mathbf{k} \cdot \mathbf{R}} \cdot \mathbf{H}_0 = -j\mathbf{k} \cdot \mathbf{H} = 0 \Rightarrow \mathbf{k} \cdot \mathbf{H} = 0\end{aligned}$$

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P8-6

The E-field of a time-harmonic uniform plane wave propagating in a dielectric medium is given by

$$\mathbf{E}(z, t) = \mathbf{a}_x 2 \cos\left(10^8 t - \frac{z}{\sqrt{3}}\right) - \mathbf{a}_y \sin\left(10^8 t - \frac{z}{\sqrt{3}}\right) \quad (\text{V/m})$$

- Determine the frequency and wavelength of the wave.
- What is the dielectric constant of the medium?
- Describe the polarization of the wave in detail.
- Find the corresponding H-field.

Solution

$\mathbf{E}(z, t)$ is a uniform plane wave propagate along z direction and is given by

$$\begin{aligned}\mathbf{E}(z, t) &= \mathcal{R}e[\mathbf{E}(z)e^{j\omega t}] \\ &= \mathcal{R}e\{[\mathbf{a}_x E_{10} e^{j\varphi_1} + \mathbf{a}_y E_{20} e^{j\varphi_2}]e^{j(\omega t - kz)}\} \\ &= \mathbf{a}_x E_{10} \cos(\omega t - kz + \varphi_1) + \mathbf{a}_y E_{20} \cos(\omega t - kz + \varphi_2),\end{aligned}$$

where E_{10} and E_{20} are real numbers denoting the amplitude of the two linearly polarized waves.

In this case, $\varphi_1 = 0$ and $\varphi_2 = \frac{\pi}{2}$

$$\begin{aligned}\mathbf{E}(z, t) &= \mathbf{a}_x E_{10} \cos(\omega t - kz) + \mathbf{a}_y E_{20} \cos(\omega t - kz + \frac{\pi}{2}) \\ &= \mathbf{a}_x E_{10} \cos(\omega t - kz) - \mathbf{a}_y E_{20} \sin(\omega t - kz)\end{aligned}$$

Solution

a) Determine the frequency and wavelength of the wave.

$$\omega = 10^8 \text{ rad/s}, k = 1/\sqrt{3}$$

$$f = \frac{\omega}{2\pi} = \frac{10^8}{2\pi} \approx 1.59 \times 10^7 \text{ Hz}$$

$$\therefore k = \frac{2\pi}{\lambda} \therefore \lambda = \frac{2\pi}{k} = 2\pi\sqrt{3} \approx 10.88 \text{ m}$$

Solution

■
b) What is the dielectric constant of the medium?

$$k = \omega \sqrt{\mu \epsilon}$$

$$\frac{1}{\sqrt{3}} = 10^8 \sqrt{\mu_0 \epsilon_0 \epsilon_r} = 10^8 \sqrt{4\pi \times 10^{-7} \times \frac{1}{36\pi} \times 10^{-9} \times \sqrt{\epsilon_r}} = \frac{1}{3} \sqrt{\epsilon_r}$$

Thus,

$$\epsilon_r = 3$$

Solution

■ c) Describe the polarization of the wave in detail.

In this case,

$$\varphi_1 = 0, \varphi_2 = \frac{\pi}{2}, E_{10} = 2, E_{20} = 1, \omega = 10^8, \text{ and } k = 1/\sqrt{3}$$

$$\begin{aligned} \mathbf{E}(z, t) &= \mathbf{a}_x E_{10} \cos(\omega t - kz + \varphi_1) + \mathbf{a}_y E_{20} \cos(\omega t - kz + \varphi_2) \\ &= \mathbf{a}_x 2 \cos(10^8 t - z/\sqrt{3} + 0) + \mathbf{a}_y 1 \cos(10^8 t - z/\sqrt{3} + \frac{\pi}{2}) \\ &= \mathbf{a}_x 2 \cos(10^8 t - z/\sqrt{3}) - \mathbf{a}_y \sin(10^8 t - z/\sqrt{3}) \end{aligned}$$

Solution

■ To observe the polarization of the wave, we set z-axis at $z = 0$.

Hence, we write

$$E_x = 2 \cos(10^8 t)$$

$$E_y = -\sin(10^8 t)$$

$$\frac{E_x^2}{E_{10}^2} + \frac{E_y^2}{E_{20}^2} = \frac{E_x^2}{2^2} + \frac{E_y^2}{(-1)^2} = \cos^2(10^8 t) + \sin^2(10^8 t) = 1$$

$$\because E_{10} \neq E_{20}$$

$\therefore$ it's elliptically polarized wave

Solution

■ And E_y leads E_x by 90° ($\frac{\pi}{2}$ rad) in time phase

Let period = T ,

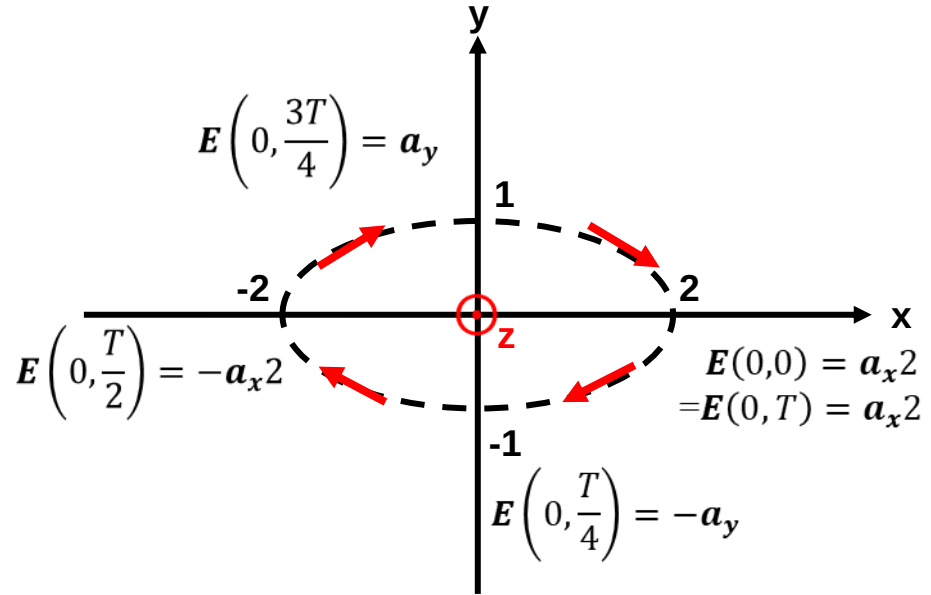
$$t = 0, \mathbf{E}(0,0) = a_x 2$$

$$t = \frac{T}{4}, \mathbf{E}\left(0, \frac{T}{4}\right) = -a_y$$

$$t = \frac{T}{2}, \mathbf{E}\left(0, \frac{T}{2}\right) = -a_x 2$$

$$t = \frac{3T}{4}, \mathbf{E}\left(0, \frac{3T}{4}\right) = a_y$$

$$t = T, \mathbf{E}(0, T) = a_x 2$$



The uniform plane wave is clockwise $\therefore$ it's left-hand elliptically polarized wave.

Solution

■ d) Find the corresponding H-field.

For propagation in + z-axis

$$\mathbf{H} = \frac{1}{\eta} \mathbf{z} \times \mathbf{E}, \quad \eta = \frac{\eta_0}{\sqrt{3}}$$

$$\mathbf{H} = \frac{\sqrt{3}}{\eta_0} \mathbf{z} \times \mathbf{E} = \frac{\sqrt{3}}{\eta_0} \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ 0 & 0 & 1 \\ 2\cos(10^8 t - z/\sqrt{3}) & -\sin(10^8 t - z/\sqrt{3}) & 0 \end{vmatrix}$$

$$\mathbf{H} = \frac{\sqrt{3}}{\eta_0} [\mathbf{a}_x \sin(10^8 t - z/\sqrt{3}) + \mathbf{a}_y 2\cos(10^8 t - z/\sqrt{3})]$$

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P.8-8

Prove the following:

- a) An elliptically polarized plane wave can be resolved into right-hand and left-hand circularly polarized waves.
- b) A circularly polarized plane wave can be obtained from a superposition of two oppositely directed elliptically polarized waves.

Solution (a)

- (a) Let the following plane waves be represented by the phasor with propagation factor e^{-jkz} omitted and E_1, E_2 are complex numbers ($E_1 \neq E_2$) :

Elliptical polarization in phasor form :

$$\mathbf{E} = \mathbf{a}_x E_1 + \mathbf{a}_y E_2$$

RHCP in phasor form :

$$\mathbf{E}_{rc} = E_{rc} (\mathbf{a}_x - \mathbf{a}_y j). \text{ Let } E_{rc} = \frac{1}{2} (E_1 + jE_2)$$

LHCP in phasor form :

$$\mathbf{E}_{lc} = E_{lc} (\mathbf{a}_x + \mathbf{a}_y j). \text{ Let } E_{lc} = \frac{1}{2} (E_1 - jE_2)$$

$$\mathbf{E}_{rc} + \mathbf{E}_{lc} = \left[\frac{1}{2} (E_1 + jE_2) (\mathbf{a}_x - \mathbf{a}_y j) \right] + \left[\frac{1}{2} (E_1 - jE_2) (\mathbf{a}_x + \mathbf{a}_y j) \right] = \mathbf{a}_x E_1 + \mathbf{a}_y E_2 = \mathbf{E}$$

Solution (b)

(b) Let the following plane waves be represented by the phasor with propagation factor e^{-jkz} omitted and E is a complex number :

RHCP: Right hand elliptical polarization Left hand elliptical polarization

$\mathbf{E}_{rc} = E(\mathbf{a}_x - \mathbf{a}_y j) = E\left(\frac{1}{2}\mathbf{a}_x - j2\mathbf{a}_y\right) + E\left(\frac{1}{2}\mathbf{a}_x + j\mathbf{a}_y\right)$

LHCP: Left hand elliptical polarization Right hand elliptical polarization

$\mathbf{E}_{lc} = E(\mathbf{a}_x + \mathbf{a}_y j) = E\left(\frac{1}{2}\mathbf{a}_x + j2\mathbf{a}_y\right) + E\left(\frac{1}{2}\mathbf{a}_x - j\mathbf{a}_y\right)$

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P.8-9

Derive the following general expression of the attenuation and phase constants for conducting media:

$$\alpha = \omega \sqrt{\frac{\mu\epsilon}{2}} \left[\sqrt{1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2} - 1 \right]^{1/2} \quad (\text{Np/m}).$$

$$\beta = \omega \sqrt{\frac{\mu\epsilon}{2}} \left[\sqrt{1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2} + 1 \right]^{1/2} \quad (\text{rad/m}).$$

Solution

For conducting media, propagation constant $\gamma = \alpha + j\beta = j\omega\sqrt{\mu\varepsilon\left(1 - j\frac{\sigma}{\omega\varepsilon}\right)}$.

$$-\gamma^2 = \beta^2 - \alpha^2 - 2j\alpha\beta = \omega^2\mu\varepsilon_c = \omega^2\mu\varepsilon\left(1 - j\frac{\sigma}{\omega\varepsilon}\right).$$

$$\beta^2 - \alpha^2 = \operatorname{Re}(-\gamma^2) = \omega^2\mu\varepsilon \quad (1)$$

$$\beta^2 + \alpha^2 = |-\gamma^2| = \omega^2\mu\varepsilon\sqrt{1 + \left(\frac{\sigma}{\omega\varepsilon}\right)^2} \quad (2)$$

From (1) and (2) we obtain

$$\alpha = \omega\sqrt{\frac{\mu\varepsilon}{2}}\left(\sqrt{1 + \left(\frac{\sigma}{\omega\varepsilon}\right)^2} - 1\right)^{1/2}, \quad \beta = \omega\sqrt{\frac{\mu\varepsilon}{2}}\left(\sqrt{1 + \left(\frac{\sigma}{\omega\varepsilon}\right)^2} + 1\right)^{1/2}.$$

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P.8-11

A 3(GHz), y-polarized uniform plane wave propagates in the + x-direction in a nonmagnetic medium having a dielectric constant 2.5 and a loss tangent 10^{-2} .

- Determine the distance over which the amplitude of the propagating wave will be cut in half.
- Determine the intrinsic impedance, the wavelength, the phase velocity, and the group velocity of the wave in the medium.
- Assuming $\mathbf{E} = \mathbf{a}_y 50 \sin(6\pi 10^9 t + \pi/3)$ (V/m) at $x = 0$, write the instantaneous expression for $\mathbf{H}$ for all t and x .

Solution (a)

(a) We assume ε'' to be independent of frequency for low-loss dielectric.

$$\Rightarrow f = 3 \times 10^9 \text{ (Hz)}, \varepsilon_r = 2.5, \tan \delta_c = \frac{\varepsilon''}{\varepsilon'} = 10^{-2},$$

$$\varepsilon_c = \varepsilon' - j\varepsilon'' = \varepsilon_r \varepsilon_0 - j\varepsilon'', c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = 3 \times 10^8 \text{ (m/s)}, \eta_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}} = 120\pi \text{ } (\Omega)$$

$$\begin{aligned} \alpha &= \frac{\omega \varepsilon''}{2} \sqrt{\frac{\mu}{\varepsilon'}} = \frac{\omega \varepsilon''}{2} \sqrt{\frac{\mu_0}{\varepsilon_r \varepsilon_0}} = \frac{\omega \varepsilon''}{2} \frac{\sqrt{\mu_0}}{\sqrt{\varepsilon_r \varepsilon_0}} \times \frac{\sqrt{\varepsilon_r \varepsilon_0}}{\sqrt{\varepsilon_r \varepsilon_0}} \\ &= \frac{\omega}{2} \left(\frac{\varepsilon''}{\varepsilon_r \varepsilon_0} \right) \sqrt{\mu_0 \varepsilon_r \varepsilon_0} = \frac{\omega}{2} \left(\frac{\varepsilon''}{\varepsilon'} \right) \frac{\sqrt{\varepsilon_r}}{c} \\ &= \frac{2\pi \times 3 \times 10^9}{2} \times 10^{-2} \times \frac{\sqrt{2.5}}{3 \times 10^8} = 0.497 \text{ (Np/m)}, \end{aligned}$$

$$e^{-\alpha x} = \frac{1}{2} \Rightarrow x = \frac{1}{\alpha} \ln(2) = 1.395 \text{ (m)}.$$

Solution (b)

(b)

$$\begin{aligned}\eta_c &= \sqrt{\frac{\mu}{\varepsilon'}} \left(1 + j \frac{\varepsilon''}{2\varepsilon'} \right) = \frac{1}{\sqrt{\varepsilon_r}} \sqrt{\frac{\mu_0}{\varepsilon_0}} \left(1 + j \frac{\varepsilon''}{2\varepsilon'} \right) = \frac{1}{\sqrt{2.5}} \times 120\pi \times \left(1 + j \frac{10^{-2}}{2} \right) \\ &= 238(1 + j0.005) = 238e^{j0.0016\pi} (\Omega),\end{aligned}$$

$$\begin{aligned}\text{Eq.(8-49): } \beta &= \omega \sqrt{\mu \varepsilon'} \left(1 + \frac{1}{8} \left(\frac{\varepsilon''}{\varepsilon'} \right)^2 \right) = \omega \sqrt{\mu_0 \varepsilon_r \varepsilon_0} \left(1 + \frac{1}{8} \left(\frac{\varepsilon''}{\varepsilon'} \right)^2 \right) = \omega \frac{\sqrt{\varepsilon_r}}{c} \left(1 + \frac{1}{8} \left(\frac{\varepsilon''}{\varepsilon'} \right)^2 \right) \\ &= (2\pi \times 3 \times 10^9) \times \frac{\sqrt{2.5}}{3 \times 10^8} \left(1 + \frac{1}{8} (10^{-2})^2 \right) = 31.6\pi \text{ (rad/m)},\end{aligned}$$

$$\lambda = \frac{2\pi}{\beta} = 0.063 \text{ (m)}, \quad u_p = \frac{\omega}{\beta} = 1.899 \times 10^8 \text{ (m/s)}$$

$$u_g = \frac{1}{\frac{d\beta}{d\omega}} = \frac{1}{\sqrt{\mu \varepsilon'} \left(1 + \frac{1}{8} \left(\frac{\varepsilon''}{\varepsilon'} \right)^2 \right)} = \frac{1}{\frac{\sqrt{\varepsilon_r}}{c} \left(1 + \frac{1}{8} \left(\frac{\varepsilon''}{\varepsilon'} \right)^2 \right)} = \frac{1}{\frac{\sqrt{2.5}}{3 \times 10^8} \left(1 + \frac{1}{8} (10^{-2})^2 \right)} = 1.897 \times 10^8 \text{ (m/s)}.$$

Solution (c)

$$(c) \quad \mathbf{E}(x, t) = \mathbf{a}_y 50e^{-0.497x} \sin\left(6\pi 10^9 t - 31.6\pi x + \frac{\pi}{3}\right)$$

$$= \mathbf{a}_y 50e^{-0.497x} \cos\left(6\pi 10^9 t - 31.6\pi x + \frac{\pi}{3} - \frac{\pi}{2}\right)$$

$$= \mathbf{a}_y 50e^{-0.497x} \cos\left(6\pi 10^9 t - 31.6\pi x - \frac{\pi}{6}\right),$$

$$\mathbf{E}(x) = \mathbf{a}_y 50e^{-0.497x} e^{j(-31.6\pi x - \frac{\pi}{6})}$$

$$\Rightarrow \mathbf{H}(x) = \frac{1}{\eta_c} \mathbf{a}_x \times \mathbf{E} = \mathbf{a}_z \frac{50e^{-0.497x} e^{j(-31.6\pi x - \frac{\pi}{6})}}{238e^{j0.0016\pi}} = \mathbf{a}_z 0.21e^{-0.497x} e^{j(-31.6\pi x - \frac{\pi}{6} - 0.0016\pi)}$$

$$= \mathbf{a}_z 0.21e^{-0.497x} e^{j(-31.6\pi x - 0.168\pi)}$$

$$\Rightarrow \mathbf{H}(x, t) = \text{Re}[\mathbf{H}(x)e^{j\omega t}] = \mathbf{a}_z 0.21e^{-0.497x} \cos(6\pi 10^9 t - 31.6\pi x - 0.168\pi) \text{ (A/m)}.$$

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P.8-13

Given that the skin depth for graphite at 100 (MHz) is 0.16 (mm), determine (a) the conductivity of graphite, and (b) the distance that a 1 (GHz) wave travels in graphite such that its field intensity is reduced by 30 (dB).

Solution

■ a) Determine the conductivity of graphite.

$$\delta = 0.16 \text{ mm at } f = 100 \text{ MHz} = 10^8 \text{ Hz}$$

Since the attenuation factor is $e^{-\alpha z}$, the amplitude of a wave will be attenuated by a factor of $e^{-1} = 0.368$ when it travels a distance $\delta = \frac{1}{\alpha}$ which is called the skin depth.

$$\delta = \frac{1}{\alpha} = \frac{1}{\sqrt{\pi f \mu \sigma}}$$

$$\sigma = \frac{1}{\pi f \mu \delta^2} = \frac{1}{\pi \times 10^8 \times 4\pi \times 10^{-7} \times (0.16 \times 10^{-3})^2} \approx 0.99 \times 10^5 \text{ (S/m)}$$

Solution

- b) Determine the distance that a 1 (GHz) wave travels in graphite such that its field intensity is reduced by 30 (dB).

at $f = 1 \text{ GHz}$,

$$\alpha = \sqrt{\pi f \mu \sigma} = \sqrt{\pi \times 10^9 \times 4\pi \times 10^{-7} \times 0.99 \times 10^5} \approx 1.98 \times 10^4 \quad (\text{Np/m})$$

Reduced by 30 dB,

$$20 \log e^{-\alpha z} = -30 \text{ dB}$$

$$e^{-\alpha z} = 10^{-\frac{3}{2}}$$

$$z = \frac{\frac{3}{2} \ln 10}{\alpha} = \frac{\frac{3}{2} \ln 10}{1.98 \times 10^4} \approx 1.75 \times 10^{-4} = 0.175 \text{ mm}$$

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P8-17

Show that the instantaneous Poynting vector of a circularly polarized plane wave propagating in a lossless medium is a constant that is independent of time and distance.

(Hint: the circularly polarized plane wave $\mathbf{E} = E_0(\mathbf{a}_x \cos \omega t + \mathbf{a}_y \sin \omega t)$)

Solution

■ Poynting Vector $\mathbf{P} = \mathbf{E} \times \mathbf{H}$ and the circularly polarized plane wave $\mathbf{E} = E_0(\mathbf{a}_x \cos \omega t + \mathbf{a}_y \sin \omega t)$

$$\mathbf{H} = \frac{1}{\eta} \mathbf{z} \times \mathbf{E} = \frac{1}{\eta} \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ 0 & 0 & 1 \\ E_0 \cos \omega t & E_0 \sin \omega t & 0 \end{vmatrix} = \frac{E_0}{\eta} (-\mathbf{a}_x \sin \omega t + \mathbf{a}_y \cos \omega t)$$

$$\mathbf{P} = \mathbf{E} \times \mathbf{H} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ E_0 \cos \omega t & E_0 \sin \omega t & 0 \\ -\frac{E_0}{\eta} \sin \omega t & \frac{E_0}{\eta} \cos \omega t & 0 \end{vmatrix} = \mathbf{a}_z \left(\frac{E_0^2}{\eta} \cos^2 \omega t + \frac{E_0^2}{\eta} \sin^2 \omega t \right)$$

$$\mathbf{P} = \frac{E_0^2}{\eta} \mathbf{a}_z (\cos^2 \omega t + \sin^2 \omega t) = \frac{E_0^2}{\eta} \mathbf{a}_z$$

$\mathbf{S}$ is constant and independent of time and distance.

Ch8: Plane Electromagnetic Waves:

Plane waves in lossy media

P8-19

From the point of view of electromagnetics, the power transmitted by a lossless coaxial cable can be considered in terms of the Poynting vector inside the dielectric medium between the inner conductor and the outer sheath. Assuming that a d-c voltage V_0 applied between the inner conductor (of radius a) and the outer sheath (of inner radius b) causes a current I to flow to a load resistance, verify that the integration of the Poynting vector over the cross-sectional area of the dielectric medium equals the power $V_0 I$ that is transmitted to the load.

(Hint: Using the Cylindrical Coordinate System.)

Solution

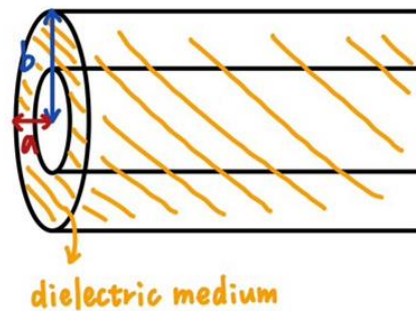
■ From Gauss's Law, $\mathbf{E} = \mathbf{a}_r \frac{\rho}{2\pi\epsilon r}$,

where ρ is the line charge density of the inner conductor.

$$V_0 = - \int_a^b \mathbf{E} \cdot d\mathbf{r} = \frac{\rho}{2\pi\epsilon} \left(- \int_a^b \mathbf{a}_r \frac{1}{r} \cdot d\mathbf{r} \right) = \frac{\rho}{2\pi\epsilon} \ln\left(\frac{b}{a}\right)$$

$$\rho = \frac{2\pi\epsilon V_0}{\ln(b/a)}$$

$$\mathbf{E} = \mathbf{a}_r \frac{V_0}{r \ln(b/a)}$$

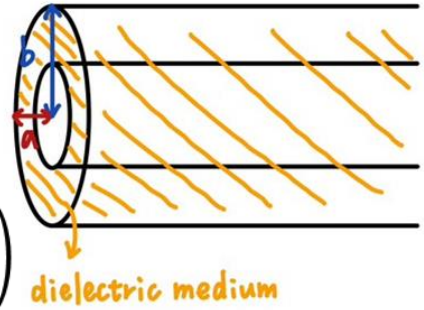


From Ampere's Circuital Law, $\mathbf{H} = \mathbf{a}_\phi \frac{I}{2\pi r}$

Solution

■ Poynting Vector $\mathbf{P} = \mathbf{E} \times \mathbf{H}$

$$\mathbf{P} = \mathbf{E} \times \mathbf{H} = \begin{vmatrix} \mathbf{a}_r & \mathbf{a}_\phi & \mathbf{a}_z \\ \frac{V_0}{r \ln(b/a)} & 0 & 0 \\ 0 & \frac{I}{2\pi r} & 0 \end{vmatrix} = \mathbf{a}_z \left(\frac{V_0 I}{2\pi r^2 \ln(b/a)} \right)$$



Power transmitted over cross-sectional area,

$$P = \oint_s \mathbf{P} \cdot d\mathbf{s} = \frac{V_0 I}{2\pi \ln(b/a)} \int_0^{2\pi} \int_a^b \left(\frac{1}{r^2} \right) r dr d\phi = V_0 I$$

Ch8: Plane Electromagnetic Waves:

Normal incidence at a plane conducting boundary

P. 8-20

A uniform plane electromagnetic wave propagates in the $+z$ -(downward) direction and impinges normally at $z = 0$ on an ocean surface. Let the magnetic field at $z = 0$ be $\mathbf{H}(0, t) = \mathbf{a}_y H_0 \cos 10^4 t$ (A/m).

- (a) Determine the skin depth. (For the ocean: Conductivity $= \sigma$, permeability $= \mu_0$)
- (b) Find the expressions for $\mathbf{H}(z, t)$ and $\mathbf{E}(z, t)$.
- (c) Find the power loss per unit area (in terms of H_0) into the ocean.

Solution (1)

(a) Use TABLE 8-1 in Example 8-4, $\frac{\sigma}{\omega\epsilon} = \frac{4}{10^4 \times (72 \times \frac{1}{36\pi} \times 10^{-9})} = 2\pi \times 10^5 \gg 1$

(The ocean is a good conductor under these conditions)

$$\Rightarrow \delta = \frac{1}{\alpha} = \frac{1}{\sqrt{\pi f \mu_0 \sigma}} = \frac{1}{\sqrt{\pi \times \frac{10^4}{2\pi} \times (4\pi \times 10^{-7}) \times 4}} = 6.31 \text{ (m)}$$

$$\begin{aligned} \text{(b) } \alpha = \beta = \sqrt{\pi f \mu_0 \sigma}, \quad \eta_c = \sqrt{\frac{\mu}{\epsilon_c}} &= \sqrt{\frac{\mu}{\epsilon(1-j\frac{\sigma}{\omega\epsilon})}} \cong \sqrt{\frac{\mu}{\epsilon(-j\frac{\sigma}{\omega\epsilon})}} = \sqrt{\frac{j\omega\mu}{\sigma}} = \sqrt{\frac{e^{j\frac{\pi}{2}}\omega\mu}{\sigma}} \\ &= \sqrt{\frac{\omega\mu}{\sigma}} e^{j\frac{\pi}{4}} = \sqrt{\frac{\omega\mu_0}{\sigma}} e^{j\frac{\pi}{4}} \quad \left(\frac{\sigma}{\omega\epsilon} = 2\pi \times 10^5 \gg 1\right) \end{aligned}$$

$$\mathbf{H}(z, t) = \mathbf{a}_y H_0 e^{-\alpha z} \cos(10^4 t - \beta z) \Rightarrow \mathbf{H}(z) = \mathbf{a}_y H_0 e^{-\alpha z} e^{-j\beta z}$$

$$\Rightarrow \mathbf{E}(z) = \eta_c \mathbf{H}(z) \times \mathbf{a}_z = \mathbf{a}_x \sqrt{\frac{\omega\mu_0}{\sigma}} e^{j\frac{\pi}{4}} H_0 e^{-\alpha z} e^{-j\beta z} = \mathbf{a}_x H_0 \sqrt{\frac{\omega\mu_0}{\sigma}} e^{-\alpha z} e^{j(-\beta z + \frac{\pi}{4})}$$

$$\Rightarrow \mathbf{E}(z, t) = \text{Re}[\mathbf{E}(z) e^{j\omega t}] = \mathbf{a}_x H_0 \sqrt{\frac{\omega\mu_0}{\sigma}} e^{-\alpha z} \cos(10^4 t - \beta z + \frac{\pi}{4})$$

Solution (2)

(c)

$$\begin{aligned}\mathbf{P}_{av} &= \frac{1}{2} \text{Re} [\mathbf{E}(z) \times \mathbf{H}(z)^*] = \frac{1}{2} \text{Re} \left[\left(\mathbf{a}_x H_0 \sqrt{\frac{\omega \mu_0}{\sigma}} e^{-\alpha z} e^{j(-\beta z + \frac{\pi}{4})} \right) \times (\mathbf{a}_y H_0 e^{-\alpha z} e^{-j\beta z})^* \right] \\&= \frac{1}{2} \text{Re} \left[\left(\mathbf{a}_x H_0 \sqrt{\frac{\omega \mu_0}{\sigma}} e^{-\alpha z} e^{j(-\beta z + \frac{\pi}{4})} \right) \times (\mathbf{a}_y H_0 e^{-\alpha z} e^{j\beta z}) \right] \\&= \frac{1}{2} \text{Re} \left[\mathbf{a}_z \left(H_0 \sqrt{\frac{\omega \mu_0}{\sigma}} e^{-\alpha z} e^{j(-\beta z + \frac{\pi}{4})} \right) (H_0 e^{-\alpha z} e^{j\beta z}) \right] \\&= \frac{1}{2} \text{Re} \left[\mathbf{a}_z \left(H_0 \sqrt{\frac{\omega \mu_0}{\sigma}} e^{-\alpha z} e^{j\frac{\pi}{4}} \right) (H_0 e^{-\alpha z}) \right] \\&= \mathbf{a}_z \frac{1}{2} H_0^2 \sqrt{\frac{\omega \mu_0}{\sigma}} e^{-2\alpha z} \text{Re} \left[e^{j\frac{\pi}{4}} \right] \\&= \mathbf{a}_z \frac{1}{2} H_0^2 \sqrt{\frac{\omega \mu_0}{\sigma}} e^{-2\alpha z} \frac{1}{\sqrt{2}} = \mathbf{a}_z \frac{H_0^2}{2} \sqrt{\frac{\omega \mu_0}{2\sigma}} e^{-2\alpha z}\end{aligned}$$

Ch8: Plane Electromagnetic Waves:

Plane waves in lossless media

Ex.8-1

A uniform plane wave with $\mathbf{E} = \mathbf{a}_x E_x$ propagates in a lossless simple medium ($\epsilon_r = 4$, $\mu_r = 1$, $\sigma = 0$) in the + z-direction. Assume that E_x is sinusoidal with a frequency 100 (MHz) and has a maximum value of + 10^{-4} (V/m) at $t = 0$ and $z = 1/8$ (m).

- Write the instantaneous expression for $\mathbf{E}$ for any t and z .
- Write the instantaneous expression for $\mathbf{H}$.
- Determine the locations where E_x is a positive maximum when $t = 10^{-8}$ (s).

Solution (a)

(a) First we find $k = \omega\sqrt{\mu\epsilon} = \omega\sqrt{\mu_r\epsilon_r}/c = \frac{2\pi 10^8 \sqrt{4}}{3 \times 10^8} = \frac{4\pi}{3}$ (rad/m)

Using $\cos \omega t$ as the reference, we find the instantaneous expression for $\mathbf{E}$ to be

$$\mathbf{E}(z, t) = \mathbf{a}_x E_x = \mathbf{a}_x 10^{-4} \cos (2\pi 10^8 t - kz + \psi)$$

Since E_x equals $+10^{-4}$ when the argument of the cosine function equals zero, that is, when $(2\pi 10^8 t - kz + \psi) = 0$, we have, at $t = 0$ and $z = 1/8$,

$$\psi = kz = \left(\frac{4\pi}{3}\right)\left(\frac{1}{8}\right) = \frac{\pi}{6} \text{ (rad)}.$$

Thus,

$$\begin{aligned} \mathbf{E}(z, t) &= \mathbf{a}_x 10^{-4} \cos \left(2\pi 10^8 t - \frac{4\pi}{3} z + \frac{\pi}{6} \right) \\ &= \mathbf{a}_x 10^{-4} \cos \left(2\pi 10^8 t - \frac{4\pi}{3} \left(z - \frac{1}{8} \right) \right) \text{ (V/m)}. \end{aligned}$$

Solution (b)

(b) The phasor expression for $\mathbf{H}$ is

$$\mathbf{H} = \mathbf{a}_y H_y = \mathbf{a}_y \frac{E_x}{\eta},$$

where

$$\eta = \sqrt{\frac{\mu}{\epsilon}} = \frac{\eta_0}{\sqrt{\epsilon_r}} = 60\pi \quad (\Omega).$$

Hence,

$$\mathbf{H}(z, t) = \mathbf{a}_y \frac{10^{-4}}{60\pi} \cos \left(2\pi 10^8 t - \frac{4\pi}{3} \left(z - \frac{1}{8} \right) \right) \quad (\text{A/m}).$$

Solution (c)

(c) At $t = 10^{-8}$, we equate the argument of the cosine function to $+2n\pi$ in order to make E_x a positive maximum:

$$2\pi 10^8(10^{-8}) - \frac{4\pi}{3} \left(z_m - \frac{1}{8} \right) = \pm 2n\pi,$$

from which we get

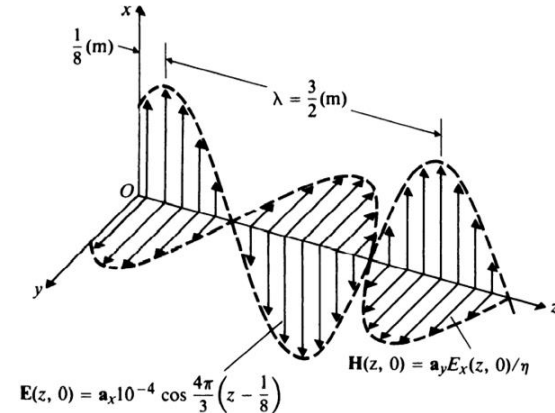
$$z_m = \frac{13}{8} \pm \frac{3}{2}n \quad (\text{m}), \quad n = 0, 1, 2, \dots; \quad z_m > 0.$$

The wavelength in the given medium is

$$\lambda = \frac{2\pi}{k} = \frac{3}{2} \quad (\text{m}).$$

Hence the positive maximum value of E_x occurs at

$$z_m = \frac{13}{8} \pm n\lambda \quad (\text{m}).$$



Ch8: Plane Electromagnetic Waves:

Plane waves in lossy media

Ex.8-4

The electric field intensity of a linearly polarized uniform plane wave propagating in the + z-direction in seawater is $\mathbf{E} = \mathbf{a}_x 100 \cos(10^7 \pi t)$ (V/m) at $z = 0$. The constitutive parameters of seawater are $\epsilon_r = 72$, $\mu_r = 1$, $\sigma = 4$ (S/m).

- Determine the attenuation constant, phase constant, intrinsic impedance, phase velocity, wavelength, and skin depth.
- Find the distance at which the amplitude of $\mathbf{E}$ is 1% of its value at $z = 0$.
- Write the expression for $\mathbf{E}(z, t)$ and $\mathbf{H}(z, t)$ at $z = 0.8$ (m) as functions of t .

Solution (a)

$$(a) \quad \omega = 10^7 \pi \text{ (rad/s)}, \quad f = \frac{\omega}{2\pi} = 5 \times 10^6 \text{ (Hz)},$$

$$\frac{\sigma}{\omega \epsilon} = \frac{\sigma}{\omega \epsilon_0 \epsilon_r} = \frac{4}{10^7 \pi \left(\frac{1}{36\pi} \times 10^{-9} \right) 72} = 200 \gg 1.$$

Hence we can use the formulas for good conductors.

Attenuation constant:

$$\alpha = \sqrt{\pi f \mu \sigma} = \sqrt{5\pi 10^6 (4\pi 10^{-7}) 4} = 8.89 \text{ (Np/m)}.$$

Phase constant:

$$\beta = \sqrt{\pi f \mu \sigma} = 8.89 \text{ (rad/m)}.$$

Solution (a)

Intrinsic impedance:

$$\eta_c = (1 + j) \sqrt{\frac{\pi f \mu}{\sigma}} = (1 + j) \sqrt{\frac{\pi(5 \times 10^6)(4\pi \cdot 10^{-7})}{4}} = \pi e^{j\pi/4} (\Omega).$$

Phase velocity:

$$u_p = \frac{\omega}{\beta} = \frac{10^7 \pi}{8.89} = 3.53 \times 10^6 \quad (\text{m/s}).$$

Wavelength:

$$\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{8.89} = 0.707 \text{ (m)}.$$

Skin depth:

$$\delta = \frac{1}{\alpha} = \frac{1}{8.89} = 0.112 \text{ (m)}.$$

Solution (b)

- (b) Distance z_1 at which the amplitude of wave decreases to 1% of its value at $z = 0$:

$$e^{-\alpha z_1} = 0.01 \text{ or } e^{\alpha z_1} = \frac{1}{0.01} = 100,$$

$$z_1 = \frac{1}{\alpha} \ln(100) = \frac{4.605}{8.89} = 0.518 \text{ (m)}.$$

Solution (c)

(c) In phasor notation,

$$\mathbf{E}(z) = \mathbf{a}_x 100 e^{-\alpha z} e^{-j\beta z}.$$

The instantaneous expression for $\mathbf{E}$ is

$$\begin{aligned}\mathbf{E}(z, t) &= \text{Re}[\mathbf{E}(z) e^{j\omega t}] = \text{Re}[\mathbf{a}_x 100 e^{-\alpha z} e^{j(\omega t - \beta z)}] \\ &= \mathbf{a}_x 100 e^{-\alpha z} \cos(\omega t - \beta z).\end{aligned}$$

At $z = 0.8$ (m) we have

$$\begin{aligned}\mathbf{E}(0.8, t) &= \mathbf{a}_x 100 e^{-0.8\alpha} \cos(10^7 \pi t - 0.8\beta) \\ &= \mathbf{a}_x 0.082 \cos(10^7 \pi t - 7.11) \text{ (V/m)}.\end{aligned}$$

Solution (c)

$$\mathbf{H}(z) = \frac{1}{\eta_c} \mathbf{a}_z \times \mathbf{E}(z) = \mathbf{a}_y \frac{100e^{-\alpha z} e^{-j\beta z}}{\eta_c} = \mathbf{a}_y \frac{100e^{-\alpha z} e^{-j\beta z}}{\pi e^{j\pi/4}} = \mathbf{a}_y \frac{100e^{-\alpha z}}{\pi} e^{-j(\beta z + \pi/4)}$$

$$\mathbf{H}(z, t) = \text{Re}[\mathbf{H}(z) e^{j\omega t}] = \text{Re}\left[\mathbf{a}_y \frac{100e^{-\alpha z}}{\pi} e^{j[\omega t - (\beta z + \frac{\pi}{4})]}\right] = \mathbf{a}_y \frac{100e^{-\alpha z}}{\pi} \cos\left(\omega t - \beta z - \frac{\pi}{4}\right)$$

$$\begin{aligned}\mathbf{H}(0.8, t) &= \mathbf{a}_y \frac{100e^{-0.8\alpha}}{\pi} \cos\left(\omega t - 0.8\beta - \frac{\pi}{4}\right) = \mathbf{a}_y \frac{100e^{-0.8 \times 8.89}}{\pi} \cos\left(10^7 \pi t - 0.8 \times 8.89 - \frac{\pi}{4}\right) \\ &= \mathbf{a}_y 0.026 \cos(10^7 \pi t - 7.897) = \mathbf{a}_y 0.026 \cos(10^7 \pi t - 7.897 + 2\pi) \\ &= \mathbf{a}_y 0.026 \cos(10^7 \pi t - 1.61) \text{ (A/m)}.\end{aligned}$$

Ch9

Theory and Applications of Transmission Lines

Ch9 Theory and Applications of Transmission Lines: General transmission-line equations

Ex 9.2

Demonstrate the analogy between the wave characteristics on a transmission line
an uniform plane waves in a lossy medium

Solution: Ex-9.2

1. Wave equations

a. Transmission line $\frac{d^2 V}{dz^2} = \gamma^2 V, \quad \gamma = \sqrt{(R + j\omega L)(G + j\omega C)}$

a. Lossy media $\frac{d^2 E}{dz^2} = \gamma^2 E, \quad \gamma = \sqrt{j\omega\mu(\sigma + j\omega\epsilon)}$

a. Both satisfy the same wave equation with complex propagation constant $\gamma = \alpha + j\beta$

1. Parameter correspondence

Transmission line	Lossy medium
Series impedance $R + j\omega L$	Magnetic term $j\omega\mu$
Shunt admittance $G + j\omega C$	Electric term $\sigma + j\omega\epsilon$

1. Characteristic Impedance

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}} \quad \eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}}$$

Ch9 Theory and Applications of Transmission Lines: General transmission-line equations

R. 9-16

What is meant by a "distortionless line"? What relation must the distributed parameters of a line satisfy in order for the line to be distortion less?

Solution R. 9-16

Meaning:

A transmission line is *distortionless* if all frequency components propagate with the **same phase velocity**, so the signal waveform is preserved (attenuation may occur, but no waveform distortion).

Condition (Heaviside condition):

The distributed parameters must satisfy $\frac{R}{L} = \frac{G}{C}$

where R is resistance per unit length, L is inductance per unit length, G is conductance per unit length, and C is capacitance per unit length.

Result:

The attenuation constant is frequency-independent, and the phase constant varies linearly with frequency.

Ch9 Theory and Applications of Transmission Lines: General transmission-line equations

Ex 9.3

It is found that the attenuation on a $50\ (\Omega)$ distortionless transmission line is $0.01\ (\text{dB/m})$. The line has a capacitance of $0.1\ (\text{nF/m})$

- (a) Find the resistance, inductance, and conductance per meter of the line.
- (b) Find the velocity of wave propagation.
- (c) Determine the percentage to which the amplitude of a voltage traveling wave decreases in $1\ (\text{km})$ and in $5\ (\text{km})$

Solution Ex 9.3

Given

$$Z_0 = 50 \, \Omega, \quad \alpha = 0.01 \, \text{dB/m}, \quad C = 0.1 \, \text{nF/m} = 1.0 \times 10^{-10} \, \text{F/m}$$

For a **distortionless line**:

$$\frac{R}{L} = \frac{G}{C}, \quad Z_0 = \sqrt{\frac{L}{C}}, \quad \alpha = \sqrt{RG}$$

Convert attenuation:

$$\alpha = \frac{\ln 10}{20} (0.01) = 1.151 \times 10^{-3} \, \text{Np/m}$$

Solution Ex 9.3 (a)

(a) R , L , G

$$L = Z_0^2 C = 50^2 (1.0 \times 10^{-10}) = 2.5 \times 10^{-7} \text{ H/m}$$

$$\sqrt{LC} = \sqrt{(2.5 \times 10^{-7})(1.0 \times 10^{-10})} = 5.0 \times 10^{-9}$$

$$k = \frac{\alpha}{\sqrt{LC}} = \frac{1.151 \times 10^{-3}}{5.0 \times 10^{-9}} = 2.30 \times 10^5$$

$$R = kL = 5.76 \times 10^{-2} \Omega/\text{m}$$

$$G = kC = 2.30 \times 10^{-5} \text{ S/m}$$

$$R = 5.76 \times 10^{-2} \Omega/\text{m}, L = 2.50 \times 10^{-7} \text{ H/m}, G = 2.30 \times 10^{-5} \text{ S/m}$$

Solution Ex 9.3 (b)

(b) Wave velocity

$$v = \frac{1}{\sqrt{LC}} = \frac{1}{5.0 \times 10^{-9}} = 2.0 \times 10^8 \text{ m/s}$$

$$v = 2.0 \times 10^8 \text{ m/s}$$

Solution Ex 9.3 (c)

(c) Amplitude decrease

$$\frac{V(z)}{V(0)} = 10^{-A_{\text{dB}}/20}$$

1 km:

$$A = 0.01 \times 1000 = 10 \text{ dB} \Rightarrow \frac{V}{V_0} = 0.316$$

Decrease = 68.4%

5 km:

$$A = 0.01 \times 5000 = 50 \text{ dB} \Rightarrow \frac{V}{V_0} = 0.00316$$

Decrease = 99.68%

1 km: 68.4% ↓, 5 km: 99.68% ↓

Ch9 Theory and Applications of Transmission Lines: General transmission-line equations

P.9-9

The following characteristics have been measured on a lossy transmission line at 100 MHz:

$$Z_0 = 50 + j0 \, (\Omega),$$

$$\alpha = 0.01 \, (\text{dB/m}),$$

$$\beta = 0.8\pi \, (\text{rad/m}).$$

Determine R , L , G , and C for the line.

Solution P-9.9

Given

$$f = 100 \text{ MHz} \Rightarrow \omega = 2\pi \times 10^8 \text{ rad/s}$$

$$Z_0 = 50 \, \Omega \text{ (purely real)}$$

$$\alpha = 0.01 \text{ dB/m} = 0.01/8.686 = 1.15 \times 10^{-3} \text{ Np/m}$$

$$\beta = 0.8\pi = 2.513 \text{ rad/m}$$

For a low-loss line:

$$Z_0 \approx \sqrt{\frac{L}{C}}, \quad \beta \approx \omega \sqrt{LC}$$

$$\alpha \approx \frac{R}{2Z_0} + \frac{GZ_0}{2}$$

Solution P-9.9

Step 1: Solve L and C

$$LC = \left(\frac{\beta}{\omega}\right)^2 = \left(\frac{2.513}{2\pi \times 10^8}\right)^2 = 1.6 \times 10^{-17}$$

$$\frac{L}{C} = Z_0^2 = 2500$$

$$\boxed{C = 80 \text{ pF/m}}, \quad \boxed{L = 200 \text{ nH/m}}$$

Step 2: Solve R and G

Assuming conductor loss dominates ($G \approx 0$):

$$R \approx 2Z_0\alpha = 2(50)(1.15 \times 10^{-3})$$

$$\boxed{R \approx 0.115 \text{ } \Omega/\text{m}}, \quad \boxed{G \approx 0 \text{ S/m}}$$

Ch9 Theory and Applications of Transmission Lines: Wave characteristics on finite transmission lines

Ex 9.5

A signal generator having an internal resistance $1\ (\Omega)$ and an open circuit voltage $v_g(t) = 0.3 \cos 2\pi 10^8 t\ (V)$ is connected to a $50\ (\Omega)$ lossless transmission line. The line is $4\ (m)$ long, and the velocity of wave propagation on the line is $2.5 \times 10^8\ (m/s)$. For a matched load, find:

- (a) the instantaneous expressions for the voltage and current at an arbitrary location on the line,
- (b) the instantaneous expressions for the voltage and current at the load,
- (c) the average power transmitted to the load.

Solution P 9.5

Given

$R_s = 1 \, \Omega$, $Z_0 = 50 \, \Omega$, $Z_L = 50 \, \Omega$ (matched, no reflection)

$$v_g(t) = 0.3 \cos(2\pi \cdot 10^8 t) \, \text{V}$$

Line length $\ell = 4 \, \text{m}$, wave velocity $v_p = 2.5 \times 10^8 \, \text{m/s}$

$$\omega = 2\pi \cdot 10^8 \, \text{rad/s}, \quad \beta = \frac{\omega}{v_p} = 0.8\pi \, \text{rad/m}, \quad \tau = \frac{\ell}{v_p} = 16 \, \text{ns}$$

Incident-wave amplitude:

$$V_0^+ = 0.3 \frac{Z_0}{R_s + Z_0} = 0.29412 \, \text{V}, \quad I_0^+ = \frac{V_0^+}{Z_0} = 5.882 \times 10^{-3} \, \text{A}$$

Solution P 9.5

1) Voltage and current at an arbitrary position z on the line

$$v(z, t) = V_0^+ \cos(\omega t - \beta z)$$

$$i(z, t) = \frac{V_0^+}{Z_0} \cos(\omega t - \beta z)$$

Numerically:

$$v(z, t) = 0.29412 \cos(2\pi \cdot 10^8 t - 0.8\pi z) \text{ V}$$

$$i(z, t) = 5.882 \times 10^{-3} \cos(2\pi \cdot 10^8 t - 0.8\pi z) \text{ A}$$

2) Voltage and current at the load ($z = \ell = 4 \text{ m}$)

$$v_L(t) = V_0^+ \cos(\omega(t - \tau))$$

$$i_L(t) = \frac{V_0^+}{Z_0} \cos(\omega(t - \tau))$$

Since $\omega\tau = 3.2\pi$:

$$v_L(t) = 0.29412 \cos(2\pi \cdot 10^8 t - 3.2\pi) \text{ V}$$

$$i_L(t) = 5.882 \times 10^{-3} \cos(2\pi \cdot 10^8 t - 3.2\pi) \text{ A}$$

3) Average power delivered to the load

$$P_{\text{avg}} = \frac{(V_0^+)^2}{2Z_0} = 8.65 \times 10^{-4} \text{ W} \approx 0.865 \text{ mW}$$

Ch9 Theory and Applications of Transmission Lines:

Wave characteristics on finite transmission lines

P. 9-14

A d-c generator of voltage V_g and internal resistance R_g is connected to a lossy transmission line characterized by a resistance per unit length R and a conductance per unit length G .

- (a) Write the governing voltage and current transmission-line equations.
- (b) Find the general solutions for $V(z)$ and $I(z)$
- (c) Specialize the solutions in part (b) to those for an infinite line.
- (d) Specialize the solutions in part (b) to those for a finite line of length l that is terminated in a load resistance R_L .

Solution P. 9-14 (a)

Assume **steady-state d-c** (so the line is characterized by **series resistance** R [Ω/m] and **shunt conductance** G [S/m]; inductance/capacitance do not enter the d-c governing equations).

Let z be measured along the line from the source toward the load.

(a) Governing transmission-line equations (d-c lossy line)

$$\boxed{\frac{dV(z)}{dz} = -R I(z)} \quad \boxed{\frac{dI(z)}{dz} = -G V(z)}$$

Combine to get second-order forms:

$$\frac{d^2 V}{dz^2} = RG V, \quad \frac{d^2 I}{dz^2} = RG I$$

Define

$$\boxed{\gamma \equiv \sqrt{RG}}, \quad \boxed{Z_0 \equiv \sqrt{\frac{R}{G}}} \text{ (“characteristic resistance”)}$$

Solution P. 9-14 (b)

(b) General solutions for $V(z)$ and $I(z)$

A convenient traveling/exponential form is

$$V(z) = V^+ e^{-\gamma z} + V^- e^{+\gamma z}$$

$$I(z) = \frac{1}{Z_0} (V^+ e^{-\gamma z} - V^- e^{+\gamma z})$$

(V^+ , V^- are constants set by boundary conditions.)

Solution P. 9-14 (c)

(c) **Infinite line** ($z \rightarrow +\infty$)

For an **infinite** line driven at $z = 0$, the solution must remain finite as $z \rightarrow \infty$, so the growing term must vanish:

$$V^- = 0$$

Thus

$$V(z) = V^+ e^{-\gamma z},$$

$$I(z) = \frac{V^+}{Z_0} e^{-\gamma z}$$

With a source V_g and internal resistance R_g ,

$$V_g = V(0) + R_g I(0) = V^+ + R_g \frac{V^+}{Z_0}$$

so

$$V^+ = \frac{V_g}{1 + R_g/Z_0} = \frac{V_g Z_0}{R_g + Z_0}$$

Therefore

$$V(z) = \frac{V_g Z_0}{R_g + Z_0} e^{-\gamma z},$$

$$I(z) = \frac{V_g}{R_g + Z_0} e^{-\gamma z}$$

Solution P. 9-14(d)

(d) Finite line of length l terminated in R_L

Boundary at the load ($z = l$):

$$V(l) = R_L I(l)$$

Define the (d-c) load reflection coefficient

$$\Gamma_L = \frac{R_L - Z_0}{R_L + Z_0}$$

and the “effective” reflection seen back at the source end

$$\Gamma_e = \Gamma_L e^{-2\gamma l}$$

Then using the general form and enforcing $V(l) = R_L I(l)$ gives

$$V(z) = V^+ (e^{-\gamma z} + \Gamma_e e^{+\gamma z})$$

$$I(z) = \frac{V^+}{Z_0} (e^{-\gamma z} - \Gamma_e e^{+\gamma z})$$

Now apply the source boundary at $z = 0$:

$$V_g = V(0) + R_g I(0)$$

with

$$V(0) = V^+ (1 + \Gamma_e), \quad I(0) = \frac{V^+}{Z_0} (1 - \Gamma_e)$$

so

$$V^+ = \frac{V_g}{(1 + \Gamma_e) + \frac{R_g}{Z_0} (1 - \Gamma_e)}$$

(Equivalent input-impedance form)

The line input resistance at $z = 0$ is

$$Z_{in} = Z_0 \frac{R_L + Z_0 \tanh(\gamma l)}{Z_0 + R_L \tanh(\gamma l)}$$

and then

$$V(0) = V_g \frac{Z_{in}}{R_g + Z_{in}}, \quad I(0) = \frac{V_g}{R_g + Z_{in}}$$

with $V(z), I(z)$ given above.

Ch9 Theory and Applications of Transmission Lines: Wave characteristics on finite transmission lines

R. 9-22 What is the input impedance of an open-circuited lossless transmission line if the length of the line is:

- (a) $\lambda/4$,
- (b) $\lambda/2$, and
- (c) $3\lambda/4$?

Solution R 9-22

For an **open-circuited, lossless transmission line** ($Z_L \rightarrow \infty$), the input impedance is

$$Z_{\text{in}} = -jZ_0 \cot(\beta l), \quad \beta = \frac{2\pi}{\lambda}.$$

- $l = \lambda/4$:

$$\beta l = \pi/2 \Rightarrow \cot(\pi/2) = 0$$

$$Z_{\text{in}} = 0 \quad (\text{short circuit})$$

- $l = \lambda/2$:

$$\beta l = \pi \Rightarrow \cot(\pi) \rightarrow \infty$$

$$Z_{\text{in}} \rightarrow \infty \quad (\text{open circuit})$$

- $l = 3\lambda/4$:

$$\beta l = 3\pi/2 \Rightarrow \cot(3\pi/2) = 0$$

$$Z_{\text{in}} = 0 \quad (\text{short circuit})$$

Ch9 Theory and Applications of Transmission Lines: Wave characteristics on finite transmission lines

Ex 9.6

The open-circuit and short-circuit impedances measured at the input terminals of a lossless transmission line of length 1.5 (m), which is less than a quarter wavelength, are $-j54.6 \text{ } (\Omega)$ and $j103 \text{ } (\Omega)$, respectively.

- (a) Find Z_0 and γ of the line.
- (b) Without changing the operating frequency, find the input impedance of a short-circuited line that is twice the given length,
- (c) How long should the short-circuited line be in order for it to appear as an open circuit at the input terminals?

Solution Ex 9.6

For a **lossless** transmission line,

$$\gamma = j\beta$$

and

$$Z_{in,oc} = -jZ_0 \cot(\beta l), \quad Z_{in,sc} = jZ_0 \tan(\beta l)$$

Given

$$Z_{in,oc} = -j54.6 \, \Omega, \quad Z_{in,sc} = j103 \, \Omega, \quad l = 1.5 \, \text{m}$$

Solution Ex 9.6

(1) Find Z_0 and γ

From magnitudes:

$$Z_0^2 = 54.6 \times 103 \Rightarrow Z_0 \approx 75.0 \, \Omega$$

$$\tan(\beta l) = \frac{103}{75.0} = 1.373 \Rightarrow \beta l = \tan^{-1}(1.373) = 0.941 \text{ rad}$$

$$\beta = \frac{0.941}{1.5} = 0.628 \text{ rad/m}$$

$$\boxed{Z_0 \approx 75.0 \, \Omega}, \quad \boxed{\gamma = j0.628 \text{ m}^{-1}}$$

(2) Input impedance of a short-circuited line of length $2l = 3 \text{ m}$

$$Z_{in} = jZ_0 \tan(2\beta l) = j(75.0) \tan(1.883) \approx -j232 \, \Omega$$

$$\boxed{Z_{in} \approx -j232 \, \Omega}$$

(3) Length for a short-circuited line to appear as an open circuit

Condition:

$$\tan(\beta L) \rightarrow \infty \Rightarrow \beta L = \frac{\pi}{2} + n\pi$$

Minimum length:

$$L = \frac{\pi}{2\beta} \approx 2.50 \text{ m}$$

$$\boxed{L_{\min} \approx 2.50 \text{ m}}, \quad \boxed{L = (2n + 1) \frac{\lambda}{4}}$$

Ch9 Theory and Applications of Transmission Lines:

Wave characteristics on finite transmission lines

R. 9-26

What is a “quarter-wave transformer”? Why is it not useful for matching a complex load impedance to a low-loss line?

Solution R 9.26

A **quarter-wave transformer** is a section of transmission line with length equal to one quarter of the guided wavelength ($\lambda/4$), inserted between a load and a main line to achieve impedance matching. For a **purely resistive load**, its characteristic impedance Z_t is chosen as

$$Z_t = \sqrt{Z_0 Z_L},$$

so that the input impedance seen looking into the transformer equals the line impedance Z_0 .

It is **not useful for matching a complex load impedance** to a low-loss line because a quarter-wave transformer can only transform **real impedances**. When the load has a reactive component, the required characteristic impedance would be complex, which is not physically realizable for a passive transmission line.

Ch9 Theory and Applications of Transmission Lines: Wave characteristics on finite transmission lines

Ex 9.8

The standing-wave ratio S on a transmission line is an easily measurable quantity.

- (a) Show how the value of a terminating resistance on a lossless line of known characteristic impedance R_0 can be determined by measuring S .
- (b) What is the impedance of the line looking toward the load at a distance equal to one quarter of the operating wavelength?

Solution Ex 9.8

On a **lossless** line with real characteristic impedance R_0 , the load reflection coefficient is

$$\Gamma_L = \frac{R_L - R_0}{R_L + R_0} \quad (R_L \text{ purely resistive})$$

and the **standing-wave ratio** (SWR) is

$$S = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|} \Rightarrow |\Gamma_L| = \frac{S - 1}{S + 1}.$$

So measuring S gives $|\Gamma_L|$. For a terminating **resistance** R_L , Γ_L is real, so there are only two possibilities:

- If $R_L > R_0$, then $\Gamma_L = +|\Gamma_L|$ and

$$R_L = R_0 \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|} = R_0 S.$$

- If $R_L < R_0$, then $\Gamma_L = -|\Gamma_L|$ and

$$R_L = R_0 \frac{1 - |\Gamma_L|}{1 + |\Gamma_L|} = \frac{R_0}{S}.$$

So **from SWR alone** you can determine

$$R_L \in \{R_0 S, R_0/S\},$$

and you pick the correct one by determining the **sign** of Γ_L (e.g., whether a nearby impedance is inductive/capacitive, or whether voltage maxima occur at the load or a quarter-wave away).

Solution Ex 9.8

Impedance a quarter-wavelength from the load

The input impedance at distance l from the load on a lossless line is

$$Z(l) = R_0 \frac{R_L + jR_0 \tan(\beta l)}{R_0 + jR_L \tan(\beta l)}.$$

At $l = \lambda/4$: $\beta l = 2\pi(\lambda/4)/\lambda = \pi/2$, so $\tan(\beta l) \rightarrow \infty$, giving the quarter-wave inversion:

$$Z(\lambda/4) = \frac{R_0^2}{R_L}.$$

(So a large resistance transforms to a small one, and vice versa.)

Ch9 Theory and Applications of Transmission Lines: Wave characteristics on finite transmission lines

Ex. 9-9

The standing-wave ratio on a lossless $50\ (\Omega)$ transmission line terminated in an unknown load impedance is found to be 3.0. The distance between successive voltage minima is 20 (cm), and the first minimum is located at 5 (cm) from the load. Determine

- (a) the reflection coefficient Γ , and
- (b) the load impedance Z_L ,

In addition, find

- (a) the equivalent length and terminating resistance of a line such that the input impedance is equal to Z_L

Solution Ex.9-9

Given: $Z_0 = 50 \Omega$, lossless line, $\text{SWR} = 3.0$, spacing between successive **voltage minima** = 20 cm, first minimum at $d = 5$ cm from the load (toward generator).

1) Reflection coefficient Γ

$$|\Gamma| = \frac{\text{SWR} - 1}{\text{SWR} + 1} = \frac{3 - 1}{3 + 1} = 0.5$$

Minima spacing = $\lambda/2 = 20$ cm $\Rightarrow \lambda = 40$ cm = 0.40 m.

$$\beta = \frac{2\pi}{\lambda} = \frac{2\pi}{0.40} = 5\pi \text{ rad/m}$$

For voltage minima: $-2\beta d + \angle\Gamma = -(2n + 1)\pi$. Using the **first** minimum ($n = 0$) at $d = 0.05$ m:

$$2\beta d = 2(5\pi)(0.05) = \frac{\pi}{2} \Rightarrow \angle\Gamma = -\pi + \frac{\pi}{2} = -\frac{\pi}{2} = -90^\circ$$

$$\Gamma = 0.5\angle-90^\circ = -j0.5$$

Solution Ex.9-9

2) Load impedance Z_L

$$Z_L = Z_0 \frac{1 + \Gamma}{1 - \Gamma} = 50 \frac{1 - j0.5}{1 + j0.5} = 50(0.6 - j0.8)$$

$$\boxed{Z_L = 30 - j40 \, \Omega}$$

3) "Equivalent" line length and terminating resistance R such that $Z_{in} = Z_L$

For a lossless line of length l terminated in a **pure resistance** R :

$$Z_{in} = Z_0 \frac{R + jZ_0 \tan(\beta l)}{Z_0 + jR \tan(\beta l)}$$

Solving for $Z_{in} = 30 + j40 \, \Omega$ gives two valid solutions (shortest positive lengths shown):

- **Solution A (shortest):** $\tan(\beta l) = +1 \Rightarrow \beta l = \pi/4$

$$l = \frac{\pi/4}{5\pi} = 0.05 \text{ m} = 5 \text{ cm}, \quad R = \frac{Z_0}{3} = 16.7 \, \Omega$$

$$\boxed{l = 5 \text{ cm}, \quad R \approx 16.7 \, \Omega}$$

- **Solution B:** $\tan(\beta l) = -1 \Rightarrow \beta l = 3\pi/4$

$$l = \frac{3\pi/4}{5\pi} = 0.15 \text{ m} = 15 \text{ cm}, \quad R = 3Z_0 = 150 \, \Omega$$

$$\boxed{l = 15 \text{ cm}, \quad R = 150 \, \Omega}$$

(Additional equivalent lengths occur by adding integer multiples of $\lambda/2 = 20 \text{ cm}$ to l .)

Ch9: Theory and Applications of Transmission Lines:

Wave characteristics of finite transmission lines

Ex.9-10

A 100(MHz) generator with $V_g = 10\angle 0^\circ$ (V) and internal resistance $50\ (\Omega)$ is connected to a lossless $50\ (\Omega)$ air line that is 3.6 (m) long and terminated in a $25 + j25\ (\Omega)$ load. Find (a) $V(z)$ at a location z from the generator, (b) V_i at the input terminals and V_L at the load, (c) the voltage standing-wave ratio on the line, and (d) the average power delivered to the load.

Solution Ex.9-10

Given

$$f = 100 \text{ MHz} \Rightarrow \lambda = 3 \text{ m}, \beta = 2\pi/\lambda = 2.094 \text{ rad/m}$$

$$Z_0 = 50 \Omega, l = 3.6 \text{ m}, V_g = 10\angle 0^\circ \text{ V}, R_s = 50 \Omega$$

$$Z_L = 25 + j25 \Omega$$

Solution Ex.9-10

(a) Voltage along the line

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} = -0.2 + j0.4$$
$$\Gamma_{in} = \Gamma_L e^{-j2\beta l} = 0.447 \angle (-27.4^\circ)$$

With $V_0^+ = 5 \angle 0^\circ$ V,

$$V(z) = V_0^+ (e^{-j\beta z} + \Gamma_{in} e^{j\beta z})$$

(b) Input and load voltages

$$V_i = 7.06 \angle (-8.4^\circ) \text{ V}$$

$$V_L = 4.47 \angle (-45.4^\circ) \text{ V}$$

(c) Voltage standing-wave ratio

$$\text{VSWR} = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|} = 2.62$$

(d) Average power delivered to the load (RMS)

$$P_L = 0.40 \text{ W}$$